

This is a part of SFQED-Loops script collection
developed for calculating loop and tree-level processes in
Strong-Field Quantum Electrodynamics.
The scripts are available on <https://github.com/ArsenyMironov/SFQED-Loops>

If you use this script in your research, please, consider
citing our papers:

- A. A. Mironov, S. Meuren, and A. M. Fedotov, PRD 102, 053005 (2020); arXiv:2003.06909
<https://doi.org/10.1103/PhysRevD.102.053005>
- A. A. Mironov, A. M. Fedotov, 105 (3), 033005 (2022); arXiv:2109.00634
<https://doi.org/10.1103/PhysRevD.105.033005>
- Y. V. Selivanov, A. A. Mironov, A. I. Alexeenko, A. M. Fedotov arXiv:2605.12479 (2026)

If you have any questions, please, don't hesitate to contact:
mironov.hep@gmail.com

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```
In[1]:= Get["FeynCalc`"]  
$PrePrint = TraditionalForm
```

FeynCalc 9.3.1 (stable version). For help, use the documentation center, check out the wiki or visit the forum.

To save your and our time, please check our FAQ for answers to some common FeynCalc questions.

See also the supplied examples. If you use FeynCalc in your research, please cite

- V. Shtabovenko, R. Mertig and F. Orellana,
Comput.Phys.Commun. 256 (2020) 107478, arXiv:2001.04407.
- V. Shtabovenko, R. Mertig and F. Orellana,
Comput.Phys.Commun. 207 (2016) 432–444, arXiv:1601.01167.
- R. Mertig, M. Böhm, and A. Denner, Comput. Phys. Commun. 64 (1991) 345–359.

Out[$\ast$]= TraditionalForm

```

In[3]:= kv[α_] = Pair[LorentzIndex[α, D], Momentum[k, D]];
av[α_] = Pair[LorentzIndex[α, D], Momentum[a, D]];
av2 = Contract[av[α] × av[α]];
Ft[α_, β_] = TensorFunction[{F, "A"},
  LorentzIndex[α, D], LorentzIndex[β, D], Dimension → D];
FFt[α_, β_] = TensorFunction[{FF, "S"},
  LorentzIndex[α, D], LorentzIndex[β, D], Dimension → D];
FDt[α_, β_] = TensorFunction[{FD, "A"},
  LorentzIndex[α, D], LorentzIndex[β, D], Dimension → D];
ScalarProduct[k, a] = 0;
ScalarProduct[k, k] = 0;
FieldSubstitutions = {Ft[α_, β_] × Ft[β_, γ_] × Ft[γ_, δ_] → 0,
  Ft[α_, β_] × Ft[β_, ε_] × Ft[δ_, ε_] → 0, Ft[α_, β_] × Ft[α_, γ_] × Ft[γ_, δ_] → 0,
  Ft[α_, β_] × Ft[α_, ε_] × Ft[δ_, ε_] → 0, Ft[α_, β_] × Ft[α_, β_] → 0,
  FDt[α_, β_] × FDt[β_, γ_] × FDt[γ_, δ_] → 0,
  FDt[α_, β_] × FDt[β_, ε_] × FDt[δ_, ε_] → 0,
  FDt[α_, β_] × FDt[α_, γ_] × FDt[γ_, δ_] → 0,
  FDt[α_, β_] × FDt[α_, ε_] × FDt[δ_, ε_] → 0, FDt[α_, β_] × FDt[α_, β_] → 0,
  FDt[α_, β_] × FDt[β_, γ_] → Ft[α, β] × Ft[β, γ],
  FDt[α_, β_] × FDt[α_, γ_] → Ft[α, β] × Ft[α, γ],
  Ft[α_, β_] × Ft[β_, γ_] → FFt[α, γ],
  Ft[α_, β_] × Ft[α_, γ_] → -FFt[β, γ],
  Ft[α_, γ_] × Ft[β_, γ_] → -FFt[α, β],
  Ft[α_, β_] × FDt[β_, γ_] → 0,
  Ft[α_, β_] × FDt[α_, γ_] → 0,
  FDt[α_, β_] × Ft[β_, γ_] → 0,
  FDt[α_, β_] × Ft[α_, γ_] → 0,
  Contract[FFt[μ_, ν_] × FFt[μ_, ν_]] → 0,
  Contract[FFt[α_, ν_] × FFt[β_, ν_]] → 0,
  Contract[Ft[μ_, ν] × kv[ν]] → 0,
  Contract[FDt[μ_, ν] × kv[ν]] → 0,
  Contract[FFt[μ_, ν] × kv[ν]] → 0,
  Contract[FFt[μ_, ν] × av[ν]] → 0,
  Contract[Ft[μ_, ν] × av[ν]] → av2 kv[μ],
  Contract[FFt[μ, ν] × kv[μ] × kv[ν]] → 0};

```

```

In[12]:= FTensor[α_, β_] = (kv[α] × av[β] - kv[β] × av[α])
FDTensor[μ_, ν_] = Contract[1 / 2 LC[μ, ν, α, β] × FTensor[α, β] /. {D → 4}]

```

Out[*]= $\alpha^\beta k^\alpha - \alpha^\alpha k^\beta$

Out[*]= $-\bar{e}^\mu \nu \bar{a} k$

```

In[14]:= NewMomentum[p_] := Module[{}],
  Print[Map[ToExpression,
    {p <> "v[α_] = Pair[LorentzIndex[α, D], Momentum[" <> p <> ", D]]",
    p <> "v2 = Contract[" <> p <> "v[α]" <> p <> "v[α]]",
    "k" <> p <> " = Contract[kv[α]" <> p <> "v[α]]",
    "F" <> p <> "v[α_] = Pair[LorentzIndex[α, D], Momentum[" <> p <> ", D]]",

```

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"FF" <> p <> "v[α_] = Pair[LorentzIndex[α, D], Momentum[FF" <> p <> ", D]]",
"FD" <> p <> "v[α_] = Pair[LorentzIndex[α, D], Momentum[FD" <> p <> ", D]]",
"a" <> p <> "=Contract[av[α]" <> p <> "v[α]]",
"ScalarProduct[k, F" <> p <> "] = 0",
"ScalarProduct[k, FF" <> p <> "] = 0",
"ScalarProduct[a, FF" <> p <> "] = 0",
"ScalarProduct[a, F" <> p <> "] = -av2 Contract[kv[μ] " <> p <> "v[μ]]",
"ScalarProduct[k, FD" <> p <> "] = 0",
"ScalarProduct[a, FD" <> p <> "] = 0",
"ScalarProduct[F" <> p <> ", F" <> p <> "] = -χ" <> p <> "^2*m^6/e^2",
"ScalarProduct[FD" <> p <> ", FD" <> p <> "] = -χ" <> p <> "^2*m^6/e^2",
"ScalarProduct[" <> p <> ", FF" <> p <> "] = χ" <> p <> "^2*m^6/e^2",
"ScalarProduct[" <> p <> ", FD" <> p <> "] = 0",
"ScalarProduct[" <> p <> ", F" <> p <> "] = 0",
"ScalarProduct[F" <> p <> ", FD" <> p <> "] = 0",
"ScalarProduct[FF" <> p <> ", FF" <> p <> "] = 0",
"ScalarProduct[F" <> p <> ", FF" <> p <> "] = 0",
"ScalarProduct[FD" <> p <> ", FF" <> p <> "] = 0"
}]]];
FieldSubstitutions = Join[FieldSubstitutions,
Map[ToExpression,
{"Contract[Ft[μ_, v] " <> p <> "v[v]] -> F" <> p <> "v[μ]",
"Contract[FDt[μ_, v] " <> p <> "v[v]] -> FD" <> p <> "v[μ]",
"Contract[FFt[μ_, v] " <> p <> "v[v]] -> FF" <> p <> "v[μ]",
"Contract[Ft[μ_, v] F" <> p <> "v[v]] -> FF" <> p <> "v[μ]",
"Contract[FDt[μ_, v] FD" <> p <> "v[v]] -> FF" <> p <> "v[μ]",
"FF" <> p <> "v[μ_] FF" <> p <> "v[v_] -> χ" <> p <> "^2*m^6/e^2 FFt[μ, v]",
"Contract[FFt[α, β] " <>
p <> "v[α] " <> p <> "v[β]] -> χ" <> p <> "^2*m^6/e^2",
"Contract[FFt[μ_, v] FF" <> p <> "v[v]] -> 0",
"Contract[FFt[μ, v] " <> p <> "v[μ] F" <> p <> "v[v]] -> 0",
"Contract[FFt[μ, v] " <> p <> "v[μ] FF" <> p <> "v[v]] -> 0",
"Contract[FDt[μ_, v] F" <> p <> "v[v]] -> 0",
"Contract[FDt[μ_, v] FF" <> p <> "v[v]] -> 0",
"Contract[FFt[μ, v] " <> p <> "v[μ] kv[v]] -> 0",
"Contract[Ft[μ_, v] FD" <> p <> "v[v]] -> 0",
"Contract[FFt[μ_, v] FD" <> p <> "v[v]] -> 0",
"Contract[FDt[μ_, v] F" <> p <> "v[v]] -> 0",
"Contract[FDt[μ_, v] FF" <> p <> "v[v]] -> 0",
"Contract[Ft[μ_, v] FF" <> p <> "v[v]] -> 0"}]
];
]
In[15]:= NewCoordinate[x_] := Module[{},
Print[Map[ToExpression,
{x <> "v[α_] = Pair[LorentzIndex[α, D], Momentum[" <> x <> ", D]]",
x <> "v2 = Contract[" <> x <> "v[α] " <> x <> "v[α]]",
"k" <> x <> " = Contract[kv[α] " <> x <> "v[α]]",

```

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"a" <> x <> " = Contract[av[α] " <> x <> "v[α]"]",
"F" <> x <> "v[α_] = Pair[LorentzIndex[α, D], Momentum[F" <> x <> ", D]]",
"FF" <> x <> "v[α_] = Pair[LorentzIndex[α, D], Momentum[FF" <> x <> ", D]]",
"FD" <> x <> "v[α_] = Pair[LorentzIndex[α, D], Momentum[FD" <> x <> ", D]]",
"ScalarProduct[k, " <> x <> "] = k" <> x,
"ScalarProduct[k, F" <> x <> "] = 0",
"ScalarProduct[k, FF" <> x <> "] = 0",
"ScalarProduct[a, FF" <> x <> "] = 0",
"ScalarProduct[a, F" <>
  x <> "] = -av2 Contract[kv[μ] " <> x <> "v[μ]"]",
"ScalarProduct[k, FD" <> x <> "] = 0",
"ScalarProduct[a, FD" <> x <> "] = 0",
"ScalarProduct[F" <>
  x <> ", F" <> x <> "] = -ξ^2 k" <> x <> "^2*m^2/e^2",
"ScalarProduct[FD" <> x <> ", FD" <> x <> "] = -ξ^2 k" <> x <> "^2*m^2/e^2",
"ScalarProduct[" <> x <> ", FF" <> x <> "] = ξ^2 k" <> x <> "^2*m^2/e^2",
"ScalarProduct[" <> x <> ", FD" <> x <> "] = 0",
"ScalarProduct[" <> x <> ", F" <> x <> "] = 0",
"ScalarProduct[F" <> x <> ", FD" <> x <> "] = 0",
"ScalarProduct[FF" <> x <> ", FF" <> x <> "] = 0",
"ScalarProduct[F" <> x <> ", FF" <> x <> "] = 0",
"ScalarProduct[FD" <> x <> ", FF" <> x <> "] = 0"
}]]];
FieldSubstitutions = Join[FieldSubstitutions,
Map[ToExpression,
{"Contract[Ft[μ_, v] " <> x <> "v[v] ]→F" <> x <> "v[μ] ",
"Contract[FDt[μ_, v] " <> x <> "v[v] ]→FD" <> x <> "v[μ] ",
"Contract[FFt[μ_, v] " <> x <> "v[v] ]→FF" <> x <> "v[μ] ",
"Contract[Ft[μ_, v] F" <> x <> "v[v] ]→FF" <> x <> "v[μ] ",
"Contract[FDt[μ_, v] FD" <> x <> "v[v] ]→FF" <> x <> "v[μ] ",
"FF" <> x <> "v[μ_] FF" <> x <> "v[v_]→ξ^2 k" <> x <> "^2*m^2/e^2 FFt[μ, v] ",
"Contract[FFt[α, β] " <> x <> "v[α] " <> x <> "v[β] ]→ξ^2 k" <> x <> "^2*m^2/e^2",
"Contract[FFt[μ_, v] F" <> x <> "v[v] ]→0",
"Contract[FFt[μ_, v] FF" <> x <> "v[v] ]→0",
"Contract[FFt[μ, v] " <> x <> "v[μ] F" <> x <> "v[v] ]→0",
"Contract[FFt[μ, v] " <> x <> "v[μ] FF" <> x <> "v[v] ]→0",
"Contract[FDt[μ_, v] F" <> x <> "v[v] ]→0",
"Contract[FDt[μ_, v] FF" <> x <> "v[v] ]→0",
"Contract[FFt[μ, v] " <> x <> "v[μ] kv[v] ]→0",
"Contract[Ft[μ_, v] FD" <> x <> "v[v] ]→0",
"Contract[FFt[μ_, v] FD" <> x <> "v[v] ]→0",
"Contract[FDt[μ_, v] F" <> x <> "v[v] ]→0",
"Contract[FDt[μ_, v] FF" <> x <> "v[v] ]→0",
"Contract[Ft[μ_, v] FF" <> x <> "v[v] ]→0"}]]];

```

```

In[16]:= (*Preparing symbolic substitutions for thruple γ-
matrix combinations and recollecting F-tensors from vectors a[μ] and k[μ]*)
TripleGamma = {DiracGamma[Momentum[a_, D], D], D].
DiracGamma[Momentum[b_, D], D].DiracGamma[Momentum[c_, D], D] →
Contract[Pair[LorentzIndex[α1, D], Momentum[a, D]] ×
Pair[LorentzIndex[α2, D], Momentum[b, D]] × Pair[LorentzIndex[α3, D],
Momentum[c, D]] (MTD[α1, α2] × GAD[α3] + MTD[α2, α3] × GAD[α1] -
MTD[α1, α3] × GAD[α2] - i LCD[β, α1, α2, α3] × GAD[β].GA[5])}]
EpsToF =
{Eps[LorentzIndex[α_, D], Momentum[a, D], Momentum[k, D], Momentum[V_, D]] →
-Contract[FDt[α, β] × Pair[LorentzIndex[β, D], Momentum[V, D]]],
Eps[LorentzIndex[α_, D], Momentum[k, D], Momentum[a, D], Momentum[V_, D]] →
Contract[FDt[α, β] × Pair[LorentzIndex[β, D], Momentum[V, D]]],
Eps[LorentzIndex[α_, D], Momentum[V_, D], Momentum[a, D], Momentum[k, D]] →
-Contract[FDt[α, β] × Pair[LorentzIndex[β, D], Momentum[V, D]]],
Eps[LorentzIndex[α_, D], Momentum[V_, D], Momentum[k, D], Momentum[a, D]] →
Contract[FDt[α, β] × Pair[LorentzIndex[β, D], Momentum[V, D]]],
Eps[LorentzIndex[α_, D], Momentum[a, D], Momentum[V_, D], Momentum[k, D]] →
Contract[FDt[α, β] × Pair[LorentzIndex[β, D], Momentum[V, D]]],
Eps[LorentzIndex[α_, D], Momentum[k, D], Momentum[V_, D], Momentum[a, D]] →
-Contract[FDt[α, β] × Pair[LorentzIndex[β, D], Momentum[V, D]]]}
akToF = {DiracGamma[Momentum[k, D], D] × Pair[Momentum[a, D], Momentum[x, D]] -
DiracGamma[Momentum[a, D], D] × Pair[Momentum[k, D], Momentum[x, D]] →
DiracGamma[Momentum[Fx, D], D],
-DiracGamma[Momentum[k, D], D] × Pair[Momentum[a, D], Momentum[x, D]] +
DiracGamma[Momentum[a, D], D] × Pair[Momentum[k, D], Momentum[x, D]] →
-DiracGamma[Momentum[Fx, D], D],
DiracGamma[Momentum[a, D], D].DiracGamma[Momentum[k, D], D] → i / 2 σF,
DiracGamma[Momentum[k, D], D].DiracGamma[Momentum[a, D], D] → -i / 2 σF}

Out[*]= {(γ·a_).(γ·b_).(γ·c_) → -i γβ.γ5 εβabc + (a·b) γ·c - (a·c) γ·b + γ·a (b·c)}

Out[*]= {εα-akV- → -FD(α, V), εα-kaV- → FD(α, V), εα-V-ak → -FD(α, V),
εα-V-ka → FD(α, V), εα-aV-k → FD(α, V), εα-kV-a → -FD(α, V)}

Out[*]= {(a·x) γ·k - γ·a (k·x) → γ·Fx, γ·a (k·x) - (a·x) γ·k → -(γ·Fx), (γ·a).(γ·k) →  $\frac{i \sigma F}{2}$ , (γ·k).(γ·a) →  $-\frac{i \sigma F}{2}$ }

In[19]:= antiTripleGamma =
{DiracGamma[LorentzIndex[β_, D], D].DiracGamma[5] × Eps[LorentzIndex[β_, D],
Momentum[a_, D], Momentum[b_, D], Momentum[c_, D]] → -i
Contract[Pair[LorentzIndex[α1, D], Momentum[a, D]] × Pair[LorentzIndex[α2, D],
Momentum[b, D]] × Pair[LorentzIndex[α3, D], Momentum[c, D]]
(MTD[α1, α2] × GAD[α3] + MTD[α2, α3] × GAD[α1] - MTD[α1, α3] × GAD[α2] -
DiracGamma[LorentzIndex[α1, D], D].DiracGamma[LorentzIndex[α2, D], D].
DiracGamma[LorentzIndex[α3, D], D])}]
Out[*]= {γβ.γ5 εβ-a-b-c- → -i ((a·b) γ·c - (a·c) γ·b + γ·a (b·c) - (γ·a).(γ·b).(γ·c))}

```

```

In[20]:= Ep[z_, p_] := {
  1 - e / 2 / Pair[Momentum[k, D], Momentum[p, D]]
    Pair[Momentum[k, D], Momentum[z, D]] × DiracSlash[k, a, Dimension → D],
  - Pair[Momentum[z, D], Momentum[p, D]] +
    e Pair[Momentum[a, D], Momentum[p, D]] / 2 / Pair[Momentum[k, D], Momentum[p, D]]
    (Pair[Momentum[k, D], Momentum[z, D]]) ^ 2 +
    e^2 av2 / 6 / Pair[Momentum[k, D], Momentum[p, D]]
    (Pair[Momentum[k, D], Momentum[z, D]]) ^ 3
}

EpC[z_, p_] := {
  1 - e / 2 / Pair[Momentum[k, D], Momentum[p, D]]
    Pair[Momentum[k, D], Momentum[z, D]] × DiracSlash[a, k, Dimension → D],
  - (- Pair[Momentum[z, D], Momentum[p, D]] +
    e Pair[Momentum[a, D], Momentum[p, D]] / 2 / Pair[Momentum[k, D],
    Momentum[p, D]] (Pair[Momentum[k, D], Momentum[z, D]]) ^ 2 +
    e^2 av2 / 6 / Pair[Momentum[k, D], Momentum[p, D]]
    (Pair[Momentum[k, D], Momentum[z, D]]) ^ 3)
}

In[22]:= ψp[z_, p_] := {
  1 - e / 2 / Pair[Momentum[k, D], Momentum[p, D]]
    Pair[Momentum[k, D], Momentum[z, D]] × DiracSlash[k, a, Dimension → D],
  - Pair[Momentum[z, D], Momentum[p, D]] +
    e Pair[Momentum[a, D], Momentum[p, D]] / 2 / Pair[Momentum[k, D], Momentum[p, D]]
    (Pair[Momentum[k, D], Momentum[z, D]]) ^ 2 +
    e^2 av2 / 6 / Pair[Momentum[k, D], Momentum[p, D]]
    (Pair[Momentum[k, D], Momentum[z, D]]) ^ 3
}

In[23]:= rtilde[μ_, s_, q_, p_] := GAD[μ] A0[s, q, p] - (e / 2)
  ( DiracSlash[a, k, Dimension → D].GAD[μ] / Pair[Momentum[k, D], Momentum[q, D]] +
    GAD[μ].DiracSlash[k, a, Dimension → D] /
    Pair[Momentum[k, D], Momentum[p, D]] ) A1[s, q, p] - e^2 av2 /
  (2 Pair[Momentum[k, D], Momentum[p, D]] × Pair[Momentum[k, D], Momentum[q, D]])
  DiracSlash[k, Dimension → D] ×
  Pair[LorentzIndex[μ, D], Momentum[k, D]] A2[s, q, p]

rtildeC[μ_, s_, q_, p_] := GAD[μ] B0[s, q, p] - (e / 2)
  ( GAD[μ].DiracSlash[k, a, Dimension → D] / Pair[Momentum[k, D], Momentum[q, D]] +
    DiracSlash[a, k, Dimension → D].GAD[μ] /
    Pair[Momentum[k, D], Momentum[p, D]] ) B1[s, q, p] - e^2 av2 /
  (2 Pair[Momentum[k, D], Momentum[p, D]] × Pair[Momentum[k, D], Momentum[q, D]])
  DiracSlash[k, Dimension → D] ×
  Pair[LorentzIndex[μ, D], Momentum[k, D]] B2[s, q, p]

```

```
In[25]:= rtilde[μ, s, q, p]
         rtildeC[μ, s, q, p]
```

$$Out[25]= -\frac{e^2 k^\mu \gamma \cdot k A_2(s, q, p)}{2 (k \cdot p) (k \cdot q)} - \frac{1}{2} e A_1(s, q, p) \left(\frac{\gamma^\mu \cdot (\gamma \cdot k) \cdot (\gamma \cdot a)}{k \cdot p} + \frac{(\gamma \cdot a) \cdot (\gamma \cdot k) \cdot \gamma^\mu}{k \cdot q} \right) + \gamma^\mu A_0(s, q, p)$$

$$Out[26]= -\frac{e^2 k^\mu \gamma \cdot k B_2(s, q, p)}{2 (k \cdot p) (k \cdot q)} - \frac{1}{2} e B_1(s, q, p) \left(\frac{(\gamma \cdot a) \cdot (\gamma \cdot k) \cdot \gamma^\mu}{k \cdot p} + \frac{\gamma^\mu \cdot (\gamma \cdot k) \cdot (\gamma \cdot a)}{k \cdot q} \right) + \gamma^\mu B_0(s, q, p)$$

```
In[27]:= chif[p_] := ToExpression["χ" <> ToString[p]]
```

```
In[28]:= ktochiSubs[p_] := Module[{f},
  f[rrr_] := ToExpression["k" <> ToString[rrr]] → chif[rrr] m^2 / ξ;
  Map[f, {p}]
]
```

```
In[29]:= ktochiSubs[p, q, g]
```

$$Out[29]= \left\{ k \cdot p \rightarrow \frac{m^2 \chi p}{\xi}, k \cdot q \rightarrow \frac{m^2 \chi q}{\xi}, g \cdot k \rightarrow \frac{m^2 \chi g}{\xi} \right\}$$

```
In[30]:= ExpandVector[p_, g_, μ_] :=
```

$$\frac{\text{chif}[g] \times \text{Pair}[\text{LorentzIndex}[\mu, D], \text{Momentum}[p, D]]}{\text{chif}[p]} - \frac{1}{m^6 \text{chif}[p]^2} e^2$$

$$\text{Pair}[\text{LorentzIndex}[\mu, D], \text{Momentum}[\text{ToExpression}["FD" <> ToString[p]], D]] \times \text{Pair}[\text{Momentum}[\text{ToExpression}["FD" <> ToString[p]], D], \text{Momentum}[g, D]] - \frac{1}{m^6 \text{chif}[p]^2}$$

$$e^2 \text{Pair}[\text{LorentzIndex}[\mu, D], \text{Momentum}[\text{ToExpression}["F" <> ToString[p]], D]] \times$$

$$\text{Pair}[\text{Momentum}[\text{ToExpression}["F" <> ToString[p]], D], \text{Momentum}[g, D]] +$$

$$\frac{1}{2 m^{12} \text{chif}[g] \text{chif}[p]^3} e^2 \text{Pair}[\text{LorentzIndex}[\mu, D],$$

$$\text{Momentum}[\text{ToExpression}["FF" <> ToString[p]], D]] (-m^8 \text{chif}[g]^2 + m^8 \text{chif}[p]^2 +$$

$$e^2 \text{Pair}[\text{Momentum}[\text{ToExpression}["FD" <> ToString[p]], D], \text{Momentum}[g, D]]^2 +$$

$$e^2 \text{Pair}[\text{Momentum}[\text{ToExpression}["F" <> ToString[p]], D], \text{Momentum}[g, D]]^2)$$

```
In[31]:= adef[q_, p_] :=
```

$$-e (\text{Pair}[\text{Momentum}[a, D], \text{Momentum}[p, D]] / \text{Pair}[\text{Momentum}[k, D], \text{Momentum}[p, D]] -$$

$$\text{Pair}[\text{Momentum}[a, D], \text{Momentum}[q, D]] / \text{Pair}[\text{Momentum}[k, D], \text{Momentum}[q, D]])$$

$$\beta \text{def}[q_, p_] := e^2 \text{av}2 / 8 * (1 / \text{Pair}[\text{Momentum}[k, D], \text{Momentum}[p, D]] -$$

$$1 / \text{Pair}[\text{Momentum}[k, D], \text{Momentum}[q, D]])$$

```
In[33]:= adefInv[q_, p_, ρ_] := -ξ^3 (chif[p] - chif[q]) / chif[p] / chif[q] ρ
```

$$\beta \text{defInv}[q_, p_] := 1 / 8 \xi^3 (\text{chif}[p] - \text{chif}[q]) / \text{chif}[p] / \text{chif}[q]$$

```
In[35]:= yArg[s_, q_, p_] := (4 β[q, p]) ^ (2 / 3) \left( \frac{s}{4 \beta[q, p]} - \frac{\alpha[q, p]^2}{64 \beta[q, p]^2} \right)
```

```
In[36]:= AnPhase[s_, q_, p_] := \frac{s \alpha[q, p]}{8 \beta[q, p]} - \frac{\alpha[q, p]^3}{192 \beta[q, p]^2}
```


$$\begin{aligned}
 \text{In[37]:= } \text{Ansubs} = & \left\{ A_0[s_-, q_-, p_-] \rightarrow \text{AiryAi}[y[s, q, p]] / (4 \beta[q, p])^{(1/3)}, \right. \\
 & B_0[s_-, q_-, p_-] \rightarrow \text{AiryAi}[y[s, q, p]] / (4 \beta[q, p])^{(1/3)}, \\
 & A_1[s_-, q_-, p_-] \rightarrow \left(\frac{\alpha[q, p]}{8 \beta[q, p]} \text{AiryAi}[y[s, q, p]] - \frac{\text{I AiryAiPrime}[y[s, q, p]]}{(4 \beta[q, p])^{(1/3)}} \right) / \\
 & (4 \beta[q, p])^{(1/3)}, B_1[s_-, q_-, p_-] \rightarrow \\
 & \left(\frac{\alpha[q, p]}{8 \beta[q, p]} \text{AiryAi}[y[s, q, p]] + \frac{\text{I AiryAiPrime}[y[s, q, p]]}{(4 \beta[q, p])^{(1/3)}} \right) / (4 \beta[q, p])^{(1/3)}, \\
 & A_2[s_-, q_-, p_-] \rightarrow \left(\left(\left(\frac{\alpha[q, p]}{8 \beta[q, p]} \right)^2 - \frac{y[s, q, p]}{(4 \beta[q, p])^{(2/3)}} \right) \text{AiryAi}[y[s, q, p]] - \right. \\
 & \left. \frac{\text{I } \alpha[q, p] \text{ AiryAiPrime}[y[s, q, p]]}{(4 \beta[q, p])^{(4/3)}} \right) / (4 \beta[q, p])^{(1/3)}, \\
 & B_2[s_-, q_-, p_-] \rightarrow \left(\left(\left(\frac{\alpha[q, p]}{8 \beta[q, p]} \right)^2 - \frac{y[s, q, p]}{(4 \beta[q, p])^{(2/3)}} \right) \text{AiryAi}[y[s, q, p]] + \right. \\
 & \left. \frac{\text{I } \alpha[q, p] \text{ AiryAiPrime}[y[s, q, p]]}{(4 \beta[q, p])^{(4/3)}} \right) / (4 \beta[q, p])^{(1/3)} \\
 & \left. \right\} \\
 \text{Out[37]= } & \left\{ A_0(s_-, q_-, p_-) \rightarrow \frac{\text{Ai}(y(s, q, p))}{2^{2/3} \sqrt[3]{\beta(q, p)}}, B_0(s_-, q_-, p_-) \rightarrow \frac{\text{Ai}(y(s, q, p))}{2^{2/3} \sqrt[3]{\beta(q, p)}}, \right. \\
 & A_1(s_-, q_-, p_-) \rightarrow \frac{\frac{\alpha(q, p) \text{Ai}(y(s, q, p))}{8 \beta(q, p)} - \frac{i \text{Ai}'(y(s, q, p))}{2^{2/3} \sqrt[3]{\beta(q, p)}}}{2^{2/3} \sqrt[3]{\beta(q, p)}}, B_1(s_-, q_-, p_-) \rightarrow \frac{\frac{\alpha(q, p) \text{Ai}(y(s, q, p))}{8 \beta(q, p)} + \frac{i \text{Ai}'(y(s, q, p))}{2^{2/3} \sqrt[3]{\beta(q, p)}}}{2^{2/3} \sqrt[3]{\beta(q, p)}}, \\
 & A_2(s_-, q_-, p_-) \rightarrow \frac{\text{Ai}(y(s, q, p)) \left(\frac{\alpha(q, p)^2}{64 \beta(q, p)^2} - \frac{y(s, q, p)}{2 \sqrt[3]{2} \beta(q, p)^{2/3}} \right) - \frac{i \alpha(q, p) \text{Ai}'(y(s, q, p))}{4 \cdot 2^{2/3} \beta(q, p)^{4/3}}}{2^{2/3} \sqrt[3]{\beta(q, p)}}, \\
 & B_2(s_-, q_-, p_-) \rightarrow \frac{\text{Ai}(y(s, q, p)) \left(\frac{\alpha(q, p)^2}{64 \beta(q, p)^2} - \frac{y(s, q, p)}{2 \sqrt[3]{2} \beta(q, p)^{2/3}} \right) + \frac{i \alpha(q, p) \text{Ai}'(y(s, q, p))}{4 \cdot 2^{2/3} \beta(q, p)^{4/3}}}{2^{2/3} \sqrt[3]{\beta(q, p)}} \left. \right\}
 \end{aligned}$$

In[38]:= **AnsSubsNoFactor** =

$$\left\{ \begin{aligned} &A_0[s_-, q_-, p_-] \rightarrow \text{AiryAi}[y[s, q, p]], \quad B_0[s_-, q_-, p_-] \rightarrow \text{AiryAi}[y[s, q, p]], \\ &A_1[s_-, q_-, p_-] \rightarrow \left(\frac{\alpha[q, p]}{8\beta[q, p]} \text{AiryAi}[y[s, q, p]] - \frac{\text{I AiryAiPrime}[y[s, q, p]]}{(4\beta[q, p])^{1/3}} \right), \\ &B_1[s_-, q_-, p_-] \rightarrow \left(\frac{\alpha[q, p]}{8\beta[q, p]} \text{AiryAi}[y[s, q, p]] + \frac{\text{I AiryAiPrime}[y[s, q, p]]}{(4\beta[q, p])^{1/3}} \right), \\ &A_2[s_-, q_-, p_-] \rightarrow \left(\left(\left(\frac{\alpha[q, p]}{8\beta[q, p]} \right)^2 - \frac{y[s, q, p]}{(4\beta[q, p])^{2/3}} \right) \text{AiryAi}[y[s, q, p]] - \frac{\text{I } \alpha[q, p] \text{ AiryAiPrime}[y[s, q, p]]}{(4\beta[q, p])^{4/3}} \right), \\ &B_2[s_-, q_-, p_-] \rightarrow \left(\left(\left(\frac{\alpha[q, p]}{8\beta[q, p]} \right)^2 - \frac{y[s, q, p]}{(4\beta[q, p])^{2/3}} \right) \text{AiryAi}[y[s, q, p]] + \frac{\text{I } \alpha[q, p] \text{ AiryAiPrime}[y[s, q, p]]}{(4\beta[q, p])^{4/3}} \right) \end{aligned} \right\}$$

Out[38]= $\{A_0(s_-, q_-, p_-) \rightarrow \text{Ai}(y(s, q, p)), B_0(s_-, q_-, p_-) \rightarrow \text{Ai}(y(s, q, p)),$

$$A_1(s_-, q_-, p_-) \rightarrow \frac{\alpha(q, p) \text{Ai}(y(s, q, p))}{8\beta(q, p)} - \frac{i \text{Ai}'(y(s, q, p))}{2^{2/3} \sqrt[3]{\beta(q, p)}}, \quad B_1(s_-, q_-, p_-) \rightarrow \frac{\alpha(q, p) \text{Ai}(y(s, q, p))}{8\beta(q, p)} + \frac{i \text{Ai}'(y(s, q, p))}{2^{2/3} \sqrt[3]{\beta(q, p)}},$$

$$A_2(s_-, q_-, p_-) \rightarrow \text{Ai}(y(s, q, p)) \left(\frac{\alpha(q, p)^2}{64\beta(q, p)^2} - \frac{y(s, q, p)}{2\sqrt[3]{2}\beta(q, p)^{2/3}} \right) - \frac{i \alpha(q, p) \text{Ai}'(y(s, q, p))}{4 \times 2^{2/3} \beta(q, p)^{4/3}},$$

$$B_2(s_-, q_-, p_-) \rightarrow \text{Ai}(y(s, q, p)) \left(\frac{\alpha(q, p)^2}{64\beta(q, p)^2} - \frac{y(s, q, p)}{2\sqrt[3]{2}\beta(q, p)^{2/3}} \right) + \frac{i \alpha(q, p) \text{Ai}'(y(s, q, p))}{4 \times 2^{2/3} \beta(q, p)^{4/3}} \}$$

In[39]:= **DiracElectronPropagatorXRepr**[x_, X_, s_] := Module[{s1},

{DiracSimplify[

Contract[DotSimplify[(1/2/m/s1 GAD[α] (Pair[Momentum[x, D], LorentzIndex[α, D]] - s1 e (kv[α] × Pair[Momentum[a, D], Momentum[x, D]] - av[α] × Pair[Momentum[k, D], Momentum[x, D]]) + s1^2/3 e^2 (-av2 Pair[Momentum[k, D], Momentum[x, D]] × kv[α])) + 1).

(1 + e s1 DiracSlash[a, k, Dimension → D]) /. {s1 → s/m^2}]]],

Exp[-I Pi/2 (D/2 - 1)] Δ^(4 - D)/2^D/Pi^(D/2) * m^(D - 1)/s^(D/2),

-s - m^2 Pair[Momentum[x, D], Momentum[x, D]]/4/s -

s/12 e^2/m^2 Contract[Pair[Momentum[x, D], LorentzIndex[α, D]] ×

FTensor[α, β] × FTensor[β, γ] × Pair[Momentum[x, D], LorentzIndex[γ, D]]] +

e Pair[Momentum[a, D], Momentum[x, D]] × Pair[Momentum[k, D], Momentum[x, D]]

] /. {av2 → -m^2 ξ^2/e^2}

]

In[40]:= **PhotonPropagator0**[z_, t_, μ_, ν_] :=

{MTD[μ, ν],

-Exp[-I Pi/2 (D/2 - 1)] Δ^(4 - D)/2^D/Pi^(D/2) * m^(D - 2)/t^(D/2),

-m^2 Pair[Momentum[z, D], Momentum[z, D]]/4/t}

```

In[41]:= PhotonPropagatorExactXRepr[z_, t_, μ_, ν_] := Module[{TFzFzμν, TFDzFDzμν},
  TFzFzμν = Expand[Contract[
    Ft[μ, α] × Pair[Momentum[z, D], LorentzIndex[α, D]] ×
    Ft[ν, β] × Pair[Momentum[z, D], LorentzIndex[β, D]] /.
    {Ft[α_, β_] → kv[α] × av[β] - kv[β] × av[α]}
  ]];
  TFDzFDzμν = Expand[Contract[
    - Pair[Momentum[z, D], Momentum[z, D]] × FFt[μ, ν] +
    FFt[μ, α] × Pair[Momentum[z, D], LorentzIndex[α, D]] ×
    Pair[Momentum[z, D], LorentzIndex[ν, D]] + FFt[ν, α] × Pair[Momentum[z, D],
    LorentzIndex[α, D]] × Pair[Momentum[z, D], LorentzIndex[μ, D]] -
    TFzFzμν - Pair[Momentum[z, D], LorentzIndex[α, D]] ×
    Pair[Momentum[z, D], LorentzIndex[β, D]] × FFt[α, β] × MTD[μ, ν] /.
    {Ft[α_, β_] → kv[α] × av[β] - kv[β] × av[α], FFt[α_, β_] → -av2 kv[α] × kv[β]}
    (* /. {av2 → -m^2 ξ^2 / e^2} *)
  ]];
  {MTD[μ, ν] * J0[m^(-2) t, ξ Pair[Momentum[k, D], Momentum[z, D]] / 2 / t] +
  (- 2 i t kv[μ] × kv[ν] / m^2 / Pair[Momentum[z, D], Momentum[k, D]] ^2 +
  e^2 TFzFzμν / m^2 / ξ^2 / Pair[Momentum[z, D], Momentum[k, D]] ^2)
  J1[m^(-2) t, ξ Pair[Momentum[k, D], Momentum[z, D]] / 2 / t] +
  (- 2 i t kv[μ] × kv[ν] / m^2 / Pair[Momentum[z, D], Momentum[k, D]] ^2 +
  e^2 TFDzFDzμν / m^2 / ξ^2 / Pair[Momentum[z, D], Momentum[k, D]] ^2)
  J2[m^(-2) t, ξ Pair[Momentum[k, D], Momentum[z, D]] / 2 / t],
  Exp[-I Pi / 2 (D / 2 - 2)] Δ^ (4 - D) / 2 ^ D / Pi ^ (D / 2) * m^ (D - 2) / t^ (D / 2) / (2 Pi),
  -m^2 Pair[Momentum[z, D], Momentum[z, D]] / 4 / t]}

```

Uncomment to see examples

```

In[42]:= (*NewMomentum["p"]
NewCoordinate["x"]
FieldSubstitutions
Expand[Contract[Ft[μ, α] xv[α] Ft[ν, β] xv[β]] /. FieldSubstitutions /.
  {Fxv[α_] → kv[α] ax - av[α] kx}] *)

```

```

In[43]:= (*run the previous example to make the next one work*)
(*Contract[DiracGamma[LorentzIndex[μ, D], D].GA[5] FDxv[μ] /.
  {FDxv[μ_] → LCD[μ, α1, α2, α3] xv[α1] kv[α2] av[α3]}]
StandardForm[%]
% /. antiTripleGamma
% /. TripleGamma *)

```

```

In[44]:= (*uncomment to see examples*)
(*NewCoordinate["o"]
NewCoordinate["0"] *)
(*Ep[x, p]
EpC[y, q] *)
(*DiracElectronPropagatorXRepr[ov, 0v, s] *)
(*PhotonPropagator0[x, t, λ, δ] *)

```

```
In[45]:= (*NewCoordinate["z"]  
PhotonPropagatorExact[z,t, $\mu$ , $\nu$ ]*)
```